

Context Recall, Worked Out by Hand

The retrieval-completeness metric — of everything the *ideal* answer needs, how much did the retrieved context actually contain? It attributes each ground-truth claim to the context and divides. Computed claim by claim.

What this document does. It defines context recall as the fraction of a gold answer’s claims that can be supported by the retrieved context, walks a worked example sentence by sentence, and explains why it is the one RAGAS metric that needs a ground-truth answer — and why it is the first thing to check when a RAG system gives incomplete answers. Numbers from Appendix A.

One of four RAGAS metrics: context recall (#28, “did we retrieve *everything needed?*”) pairs with context precision (#27, “did we rank it well?”) to diagnose the retriever.

1 What it measures

Faithfulness (#25) checks the answer against the context; context recall checks the context against the *truth*. Given a ground-truth (ideal) answer, it asks: could the retrieved context support every claim that ideal answer makes? If the context is missing a fact the correct answer requires, the LLM never had a chance to include it — a silent retrieval failure that no amount of generator tuning can fix.

Intuition

Imagine the perfect answer is a recipe with five ingredients. Context recall checks your pantry (the retrieved chunks): do you have all five? If you are missing one, the dish *cannot* be made correctly no matter how good the cook (the LLM) is. So context recall measures the *ceiling* on answer completeness. It is the only one of the four metrics that requires a labelled ground-truth answer, because you can only check “did we retrieve everything needed?” if you know what “everything” is.

2 The formula

Decompose the ground-truth answer into claims (sentences). Each claim is *attributable* if it can be inferred from the retrieved context:

$$\text{Context Recall} = \frac{\#\{\text{ground-truth claims attributable to the context}\}}{\#\{\text{total ground-truth claims}\}} \in [0, 1].$$

1.0 means the retrieved context covers everything the ideal answer needs; lower means the retriever left required facts on the table.

3 Worked by hand

Question. “Tell me about the Eiffel Tower’s construction and size.”

Ground-truth answer (5 claims).

ground-truth claim	in retrieved context?	note
1. Located in Paris	yes	“in Paris, France”
2. Completed in 1889	yes	“Completed in 1889”
3. 330 metres tall	yes	“stands 330 metres tall”
4. Made of wrought iron	yes	“wrought-iron lattice”
5. Designed by Gustave Eiffel’s firm	no	context omits the designer

Score.

$$\text{Context Recall} = \frac{4 \text{ attributable}}{5 \text{ total}} = \boxed{0.80}.$$

Four of the five required facts are present; the designer claim cannot be supported because the retriever never fetched a chunk mentioning it. The best possible answer this context allows is 80% complete — a hard ceiling set entirely by retrieval.

Mini-example · why a perfect generator can’t save you

Even a flawless, perfectly faithful LLM working from this context can only produce 4 of the 5 claims — claim 5 is simply not retrievable from what it was given. Faithfulness might read a perfect 1.0 (everything the model said was grounded), yet the answer is incomplete. **Context recall is the metric that exposes this:** the failure is upstream, in retrieval, and the fix is better retrieval (more chunks, better embeddings, hybrid search) — not a better prompt.

4 Reading it alongside the others

- **Low context recall** $\Rightarrow$ the *retriever* missed needed information. Increase K , improve the embedding model, add hybrid/BM25 (Track 3), or fix chunking (#13) so the fact isn’t split or buried.
- **High recall, low faithfulness** $\Rightarrow$ retrieval was fine but the generator strayed — the context had the facts; the model invented anyway (#25).
- **The diagnostic pair:** context precision (#27) asks “well-ranked?”, context recall asks “complete?”. Together they fully characterise retrieval quality before generation even begins.

Equation cheat-sheet

$$\text{Context Recall} = \frac{\text{ground-truth claims supported by context}}{\text{total ground-truth claims}} \in [0, 1]$$

needs a ground-truth answer; low $\Rightarrow$ retriever missed needed info (ceiling on completeness)

Appendix A · Reference sketch

An LLM judge attributes each ground-truth claim to the retrieved context.

```
gt_claims = [
    ("located in Paris",      True),
    ("completed in 1889",    True),
    ("330 metres tall",      True),
```

```
    ("made of wrought iron",    True),  
    ("designed by Eiffel's firm", False),    # not in retrieved context  
]  
context_recall = sum(ok for _, ok in gt_claims) / len(gt_claims)    # 4/5 = 0.80
```

Internal learning material. Numbers reproducible from the appendix sketch. v1.0